

Image Down Sampling and Up Sampling

(Assignment 1)

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I. Introduction

Image sampling is one of the basic concepts in digital image processing. It is the process of changing the resolution of an image by either reducing or increasing it. In this assignment, three downsampling methods, which are Max Downsampling, Average Downsampling, and Median Downsampling, and three upsampling methods, which are Nearest Neighbour (NN), Bilinear, and Bicubic, were tested on three random photos. The purpose of this assignment is to see how each method changes the image and to compare them.

II. Photos Used

Three different stock photos were used as the input images, and each image was resized to 256×256 pixels before processing. The photos had different characteristics, such as colors, textures, and details. Comparing different types of images shows how each sampling method performs depending on the characteristics of the photo.

III. Downsampling

Max Downsampling reduces the image by selecting the highest pixel value in each block. This makes the image look brighter because darker pixels are often removed. It still keeps bright areas of the photo, but some details and darker areas may be lost. Therefore, Max Downsampling is more suitable when the main goal is to highlight bright features.

Average Downsampling calculates the average pixel value in each block. It gives the most balanced and natural result. However, because the pixels are averaged, some sharp edges and small details can become blurry. Overall, it is a good choice for general image downsampling.

Median Downsampling selects the middle pixel value after sorting the values in each block. This method is less affected by unusual pixels. It also preserves edges better than Average Downsampling. Because of this, Median Downsampling is more efficient for images that contain noise or small unwanted artifacts.

IV. Upsampling

Nearest Neighbour (NN) increases the image size by copying the value of the closest pixel. It is the fastest method, but the result usually still looks pixelated, especially the edges. It is very useful when speed is more important than image quality.

Bilinear uses the four closest pixels to calculate the value of each new pixel. This makes the image smoother and removes most of the blocky appearance from Nearest Neighbour.

However, the result can look slightly blurry because nearby pixels are averaged together.

Bicubic uses more surrounding pixels to create a smoother and sharper result. It usually gives the most natural looking image and keeps edges and textures better. However, it requires more computation and can sometimes create small visual errors around very sharp edges.

V. Conclusion

From this assignment, we can conclude that downsampling always reduces image information, which means some details will be lost and cannot be fully recovered even after upsampling. Each method has its own advantages depending on the type of picture. For downsampling, Average Downsampling produces the most balanced and natural result, while Median Downsampling performs better when the image contains noise or visual errors. For upsampling, Bicubic gives the sharpest and most natural result, while Bilinear is a solid choice when both quality and speed matter. Based on the results obtained, the best overall method for maintaining image quality is the combination of Average Downsampling and Bicubic Upsampling.